COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

1 Complexity Theory (ad260)

Recall that a (*simple, undirected*) graph G is a set of vertices V along with a set of edges E, where each edge $e \in E$ is a two-element subset of V. For the purpose of this question, all graphs are simple, undirected graphs.

For a graph $G = (V, E)$ and an edge $e \in E$, we write $G - e$ to denote the graph obtained from G by *removing* the edge e. That is $G - e$ has exactly the same vertices as G and all edges in E except for e.

NP-complete Problems

(*a*) What is a *Hamiltonian cycle* in a graph $G = (V, E)$? [2 marks]

Answer: A Hamiltonian cycle in G is an enumeration $v_1, v_2, \ldots, v_n$ of V such that for all i, $\{v_i, v_{i+1}\} \in E$ and $\{v_1, v_n\} \in E$.

NP-complete Problems

(*b*) What is known about the complexity of deciding whether a given graph G has a Hamiltonian cycle? [2 marks]

Answer: The problem is known to be NP-complete. In particular, this implies that no polynomial time algorithm is known. If such an algorithm were found it would imply that P=NP.

Graph problems

(*c*) Show that G has a Hamiltonian cycle that does not include the edge e if, and only if, $G - e$ has a Hamiltonian cycle. [4 marks]

Answer: Suppose first that G has a Hamiltonian cycle that does not include e. That is to say there is an enumeration $v_1, v_2, \ldots, v_n$ of V which is a Hamiltonian cycle but e is not any of the edges $\{v_i, v_{i+1}\}$ or $\{v_n, v_1\}$. But then, this is still a Hamiltonian cycle in $G - e$. Conversely, if $v_1, v_2, \ldots, v_n$ is a Hamiltonian cycle in $G - e$, it is clearly a Hamiltonian cycle of G. Moreover, e cannot be one of the edges $\{v_i, v_{i+1}\}$ or $\{v_n, v_1\}$ as it is not an edge of $G - e$, so if it were one of $\{v_i, v_{i+1}\}$ or $\{v_n, v_1\}$, this enumeration would not be a valid Hamiltonian cycle in $G - e$.

Self-reducibility of NP-complete problems

(*d*) Assume that P=NP. Using this assumption, show that there is a polynomial-time algorithm A such that if A is given a graph $G = (V, E)$, it will return "no" if G does not contain a Hamiltonian cycle and return a Hamiltonian cycle of G otherwise. [12 marks]

Answer: Since the problem of deciding whether a graph G has a Hamiltonian cycle is in the complexity class NP, if P=NP, then there is a polynomial time algorithm which decides this problem. Let us write Ham(G) for a call to this algorithm. We can now write the algorithm A as follows.

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If not Ham(G) return "no"
Let e_1,...,e_m be an enumeration of all the edges of G
For i: = 1 to m do
  if Ham(G - e_i) then
        G := G - e_i
od
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Let v_1 be any vertex of G and v_2 be a neighbour of v_1 in G.
For i := 2 to n-1
  let v_{i+1} be the unique neighbour of v_i that is different from v_{i-1}
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This algorithm runs in polynomial time by the assumption that $\mathrm{Ham}(G)$ runs in polynomial time. The algorithm contains $m = |E|$ calls to this function, each on a graph with the same number of vertices and no more edges than G, so the running time is polynomial in the size of G.

To see the correctness, note that the graph G at the end of the first loop must contain the edges of a Hamiltonian cycle and no others. Thus, enumerating the vertices in the order specified is a valid enumeration of a Hamiltonian cycle.
